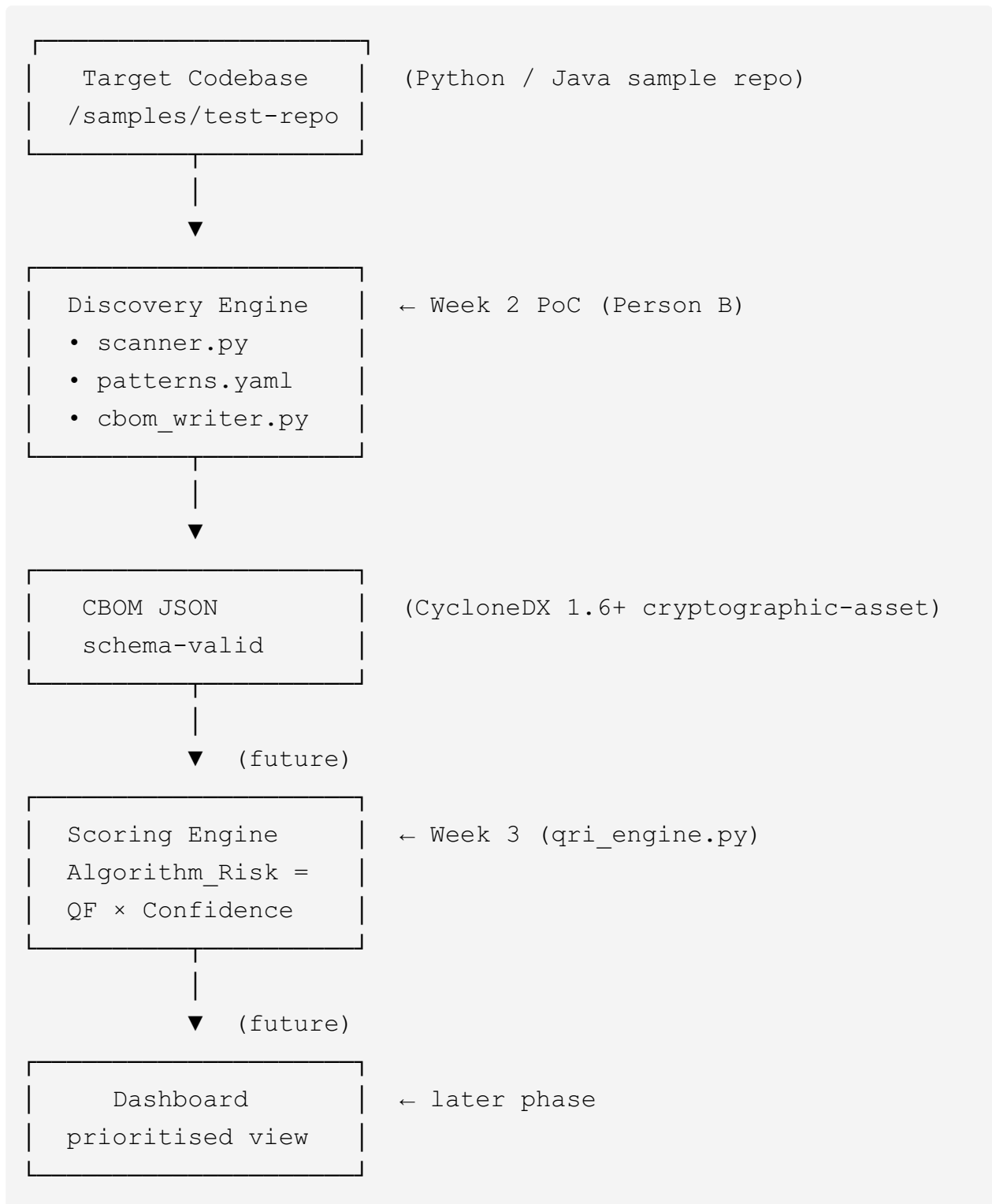


QuantumShield — High-Level Architecture



Pillar Mapping

Pillar	Component	Status
1. Discovery	scanner + CBOM writer	In progress (Week 2)
2. Scoring	QRI / Algorithm Risk Score	Spec ready, code in Week 3
3. Dashboard	Visualisation & prioritisation	Future

Data Flow Summary

1. Scanner walks directory → matches regexes from `patterns.yaml`.
2. Each match becomes a finding (file, line, keyword, confidence).
3. `cbom_writer.py` turns findings into CycloneDX `cryptographic-asset` components following `/docs/cbom-field-reference.md`.
4. Resulting `cbom.json` is the single source of truth for downstream scoring and (later) the dashboard.
5. Scoring engine reads the CBOM, applies the formula from `/docs/qri-formula-draft.md`, and emits prioritised risk records.

Design Principles

- **Schema-first:** everything downstream consumes valid CBOM.
- **Separation of concerns:** discovery never calculates risk; scoring never scans source.
- **Replaceable detectors:** the regex scanner can later be swapped for sonar-cryptography or cdxgen without changing the scoring or dashboard contracts.